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Title: Integrating Nonproliferation and Safeguards Concepts into Reactor Lab Courses

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Integrating Nonproliferation and Safeguards Concepts into Reactor Lab Courses

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NEN-5

Systems Design & Analysis

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Katie Mummah, NEN-5

- Educational Background
  - BS Nuclear, Plasma, and Radiological Engineering, University of Illinois, 2017
  - MS/PhD track, Nuclear Engineering, University of Wisconsin

- NEN-5, Systems Design & Analysis
  - Dave Poston

- Research
  - Analysis of nuclear thermal rockets
  - Nonproliferation education
  - Fuel cycle simulators

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Most undergraduate nuclear engineering students are not exposed to technical aspects of nonproliferation

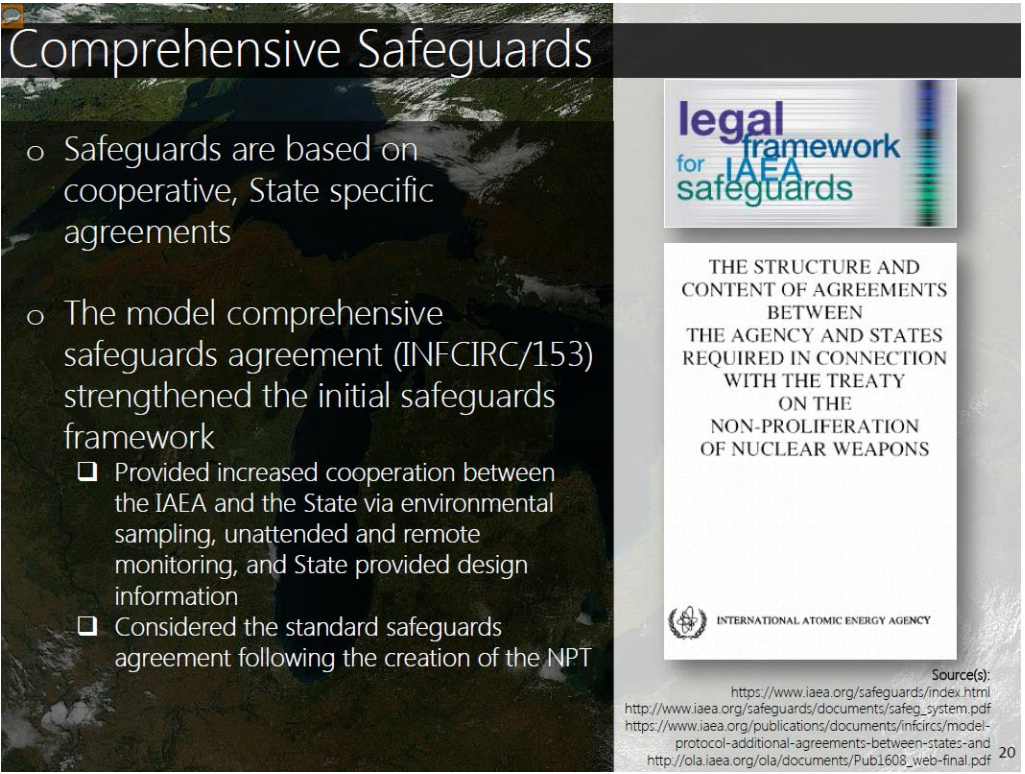
Source: NRC

Source: LA-UR-17-21214

Integral Nonproliferation Introductory Teaching & Learning (INITIAL) module integrates into existing nuclear engineering classes

Source: LA-UR-17-21214

Integral Nonproliferation Introductory Teaching & Learning (INITIAL) module integrates into existing nuclear engineering classes



Source: LA-UR-17-21214

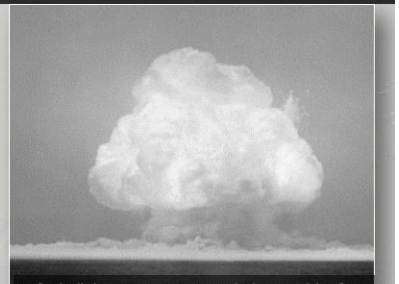
Integral Nonproliferation Introductory Teaching & Learning (INITIAL) module integrates into existing nuclear engineering classes

Comprehensive Safeguards

- Safeguards are based on cooperative, State specific agreements
- The model comprehensive safeguards agreement (INF strengthened the initial safe framework
  - Provided increased cooperatior the IAEA and the State via envir sampling, unattended and remc monitoring, and State provided information
  - Considered the standard safegu agreement following the creatic

History of the Bomb

- The first test took place at 5:30am on July 16, 1945 at the Trinity Test Site
  - The success of the Trinity Test presented a way to end the war against Japan without an invasion or help from the Soviets
- The final decision to drop the bomb was made on July 25
  - "Little Boy" an HEU-based assembly was dropped on Hiroshima August 6, 1945
  - "Fat Man" – a plutonium implosion assembly was dropped on Nagasaki August 9, 1945
  - Japan surrendered on August 10, 1945
- Tens of thousands of warheads of over 60 variations were built from 1945-1990



A fireball begins to rise, and the world's first atomic mushroom cloud begins to form, nine seconds after Trinity detonated on July 16, 1945.

Fat man in transit days before it was dropped on Nagasaki in 1945.

Source(s):
https://www.osti.gov/opennet/manhattan-project-history/Events/1942/1942.htm
https://www.TheAtlantic.com

14

Source: LA-UR-17-21214

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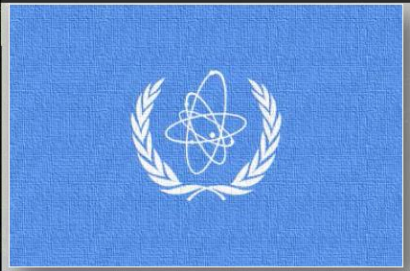
The NPT

- The NPT has Three Pillars:
  - Stop the further spread of nuclear weapons
  - Provide a sound basis for international cooperation in peaceful uses of nuclear energy
  - Commit all parties to undertake negotiations in good faith on disarmament
  - Intended to provide stability and country behavior predictability
- 62 States signed the Nuclear Non-Proliferation Treaty (NPT) on July 1, 1968
- Review Conferences are held every 5 years since to ensure that the provisions of the treaty are being properly realized
- The DPRK is the only nation to have exercised the right to withdraw provided in Article X



Source(s):
http://fas.org/nuke/control/npt/

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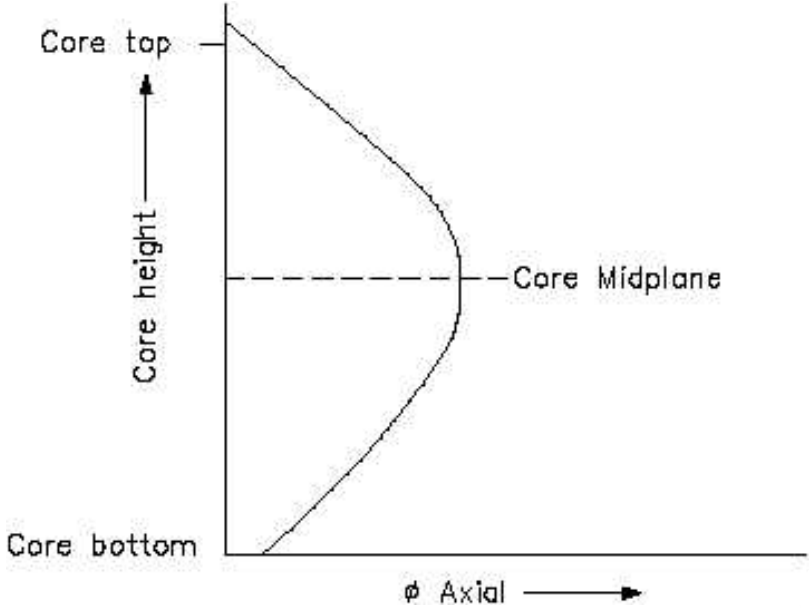

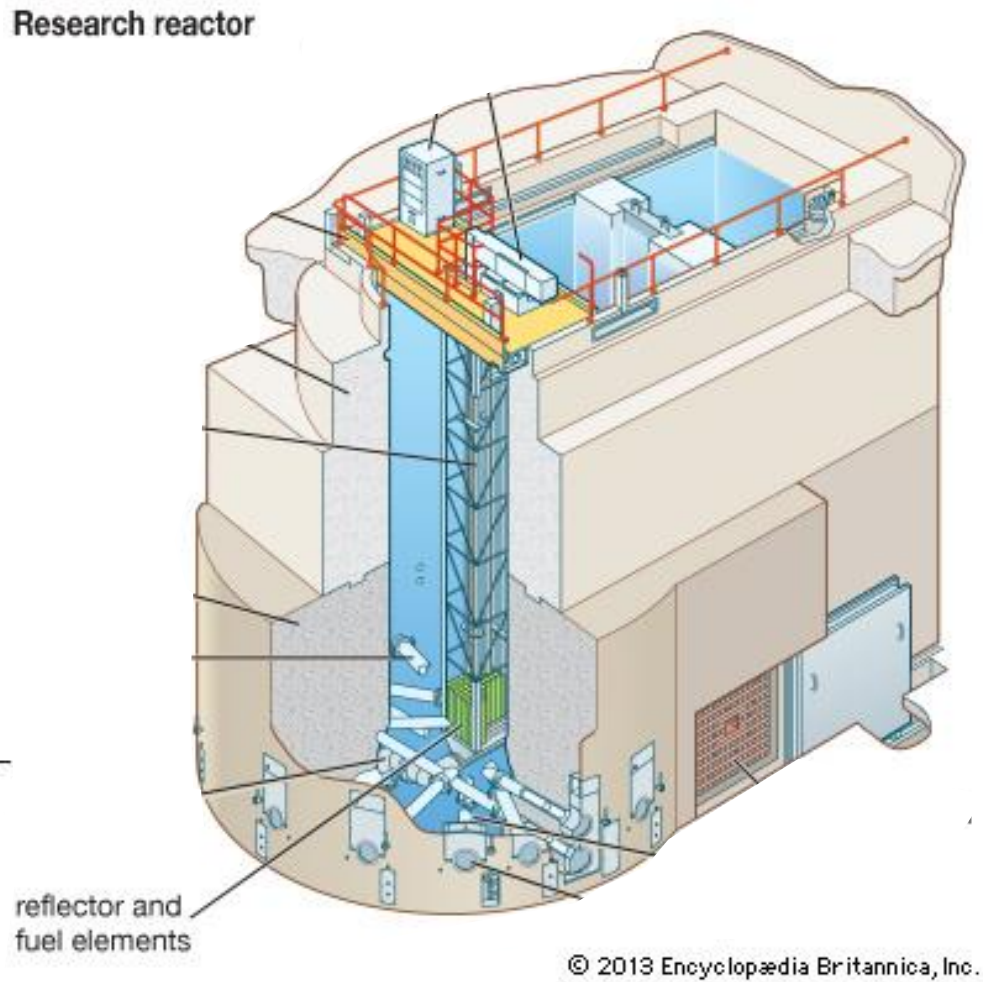

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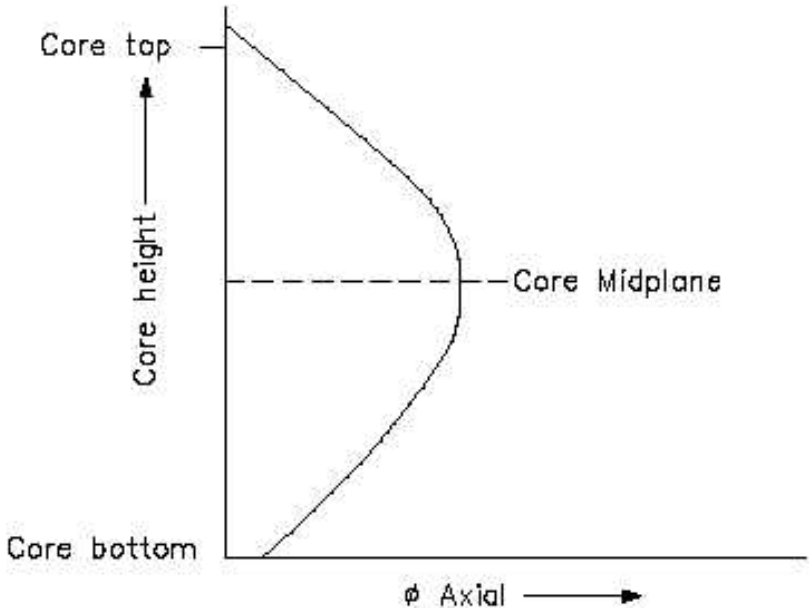

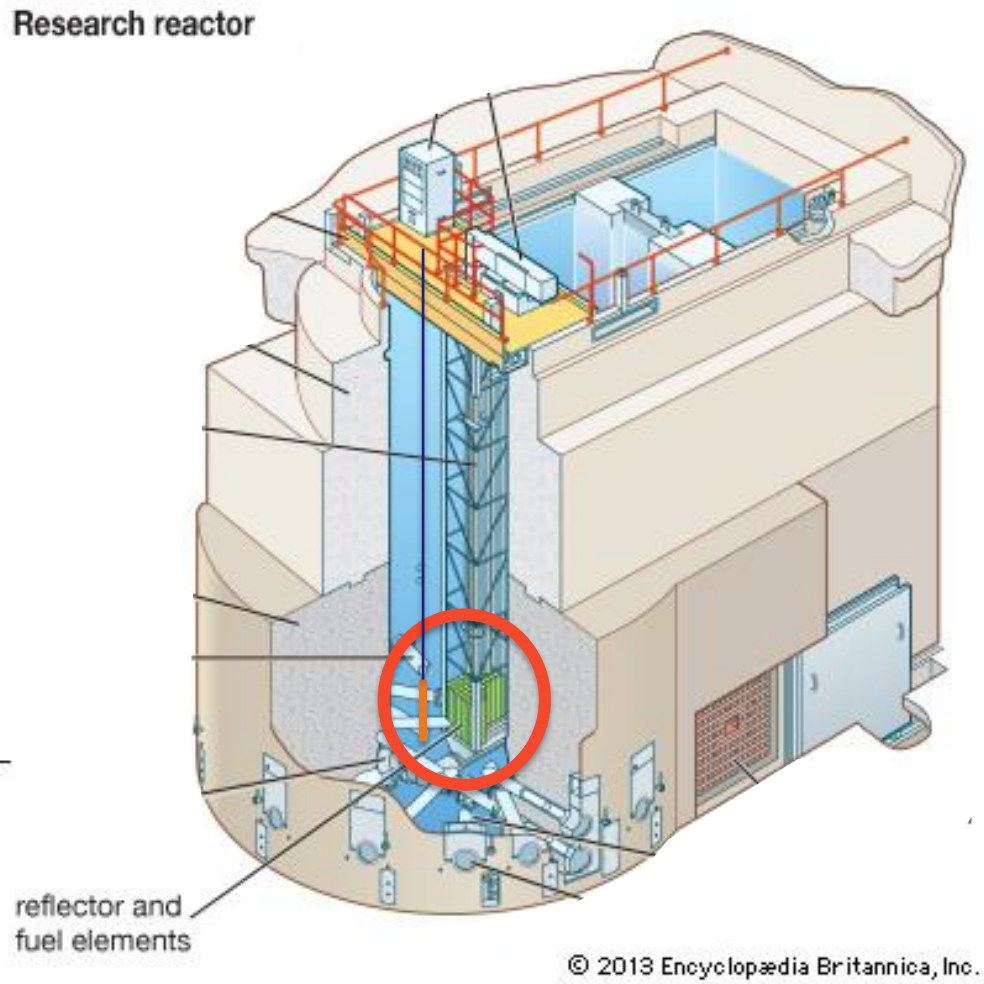

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Bottom of core

Top of core

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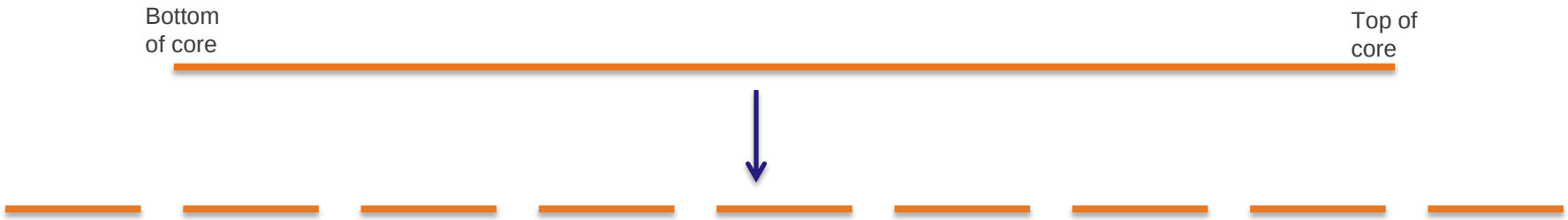

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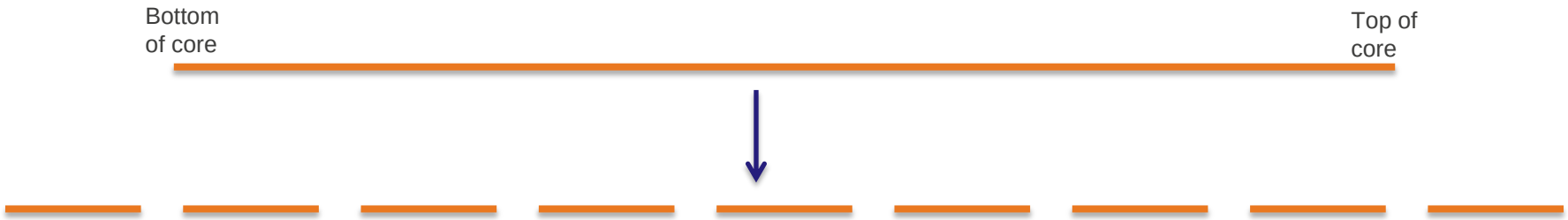

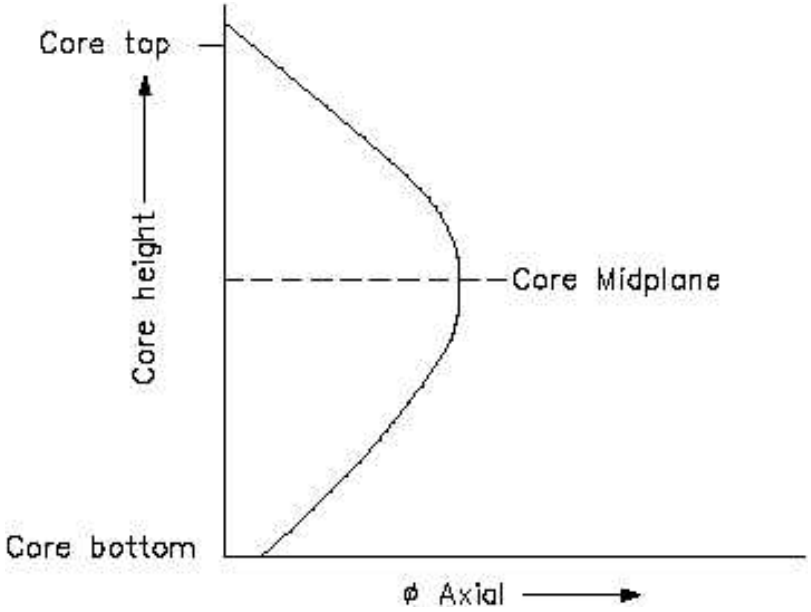

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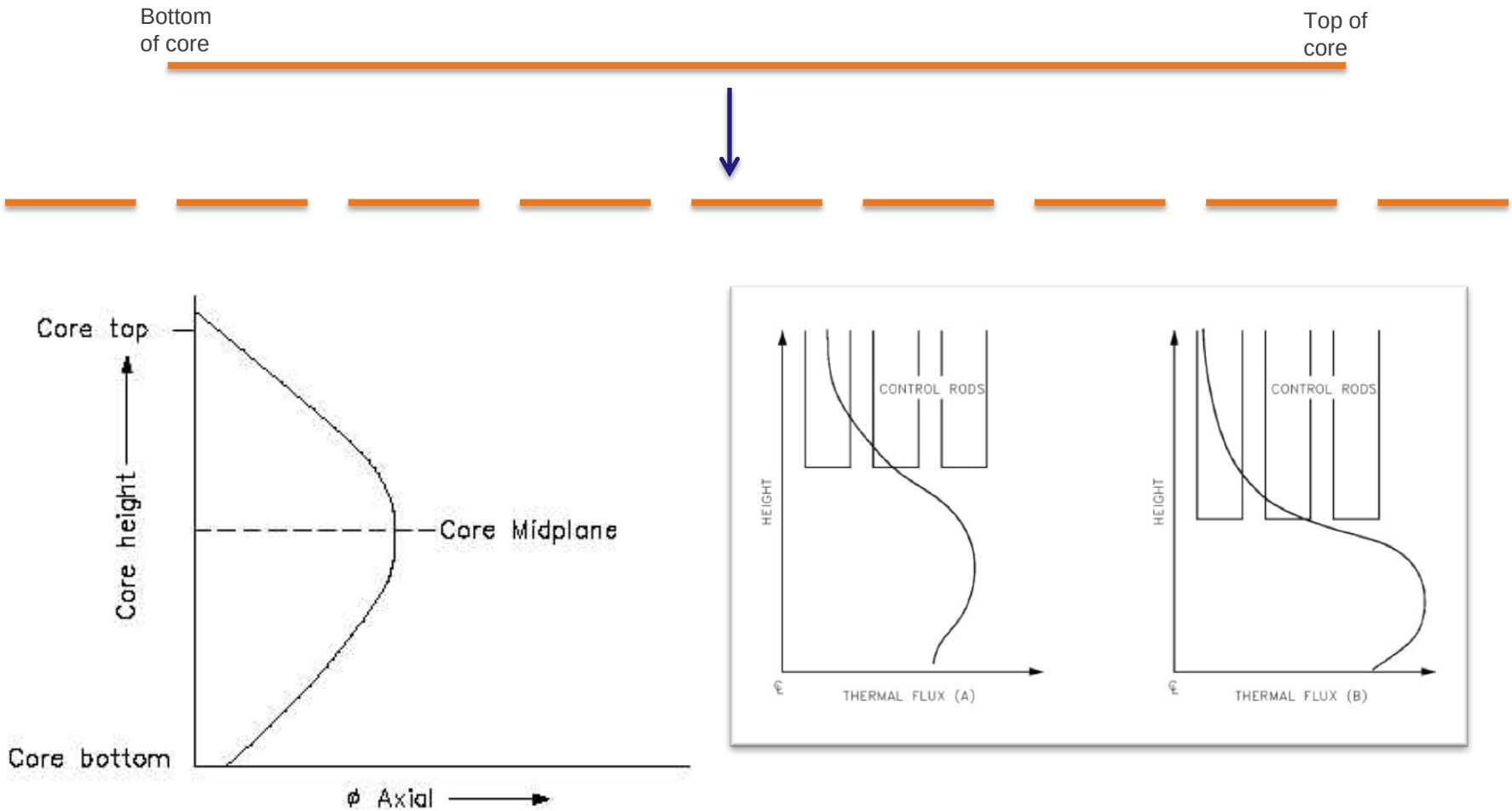

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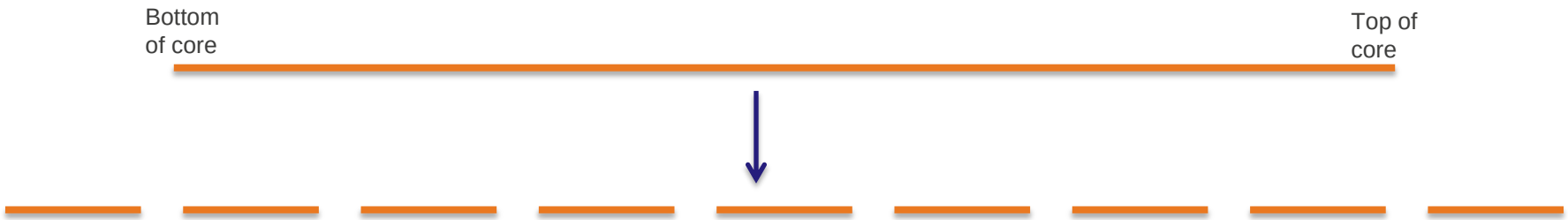

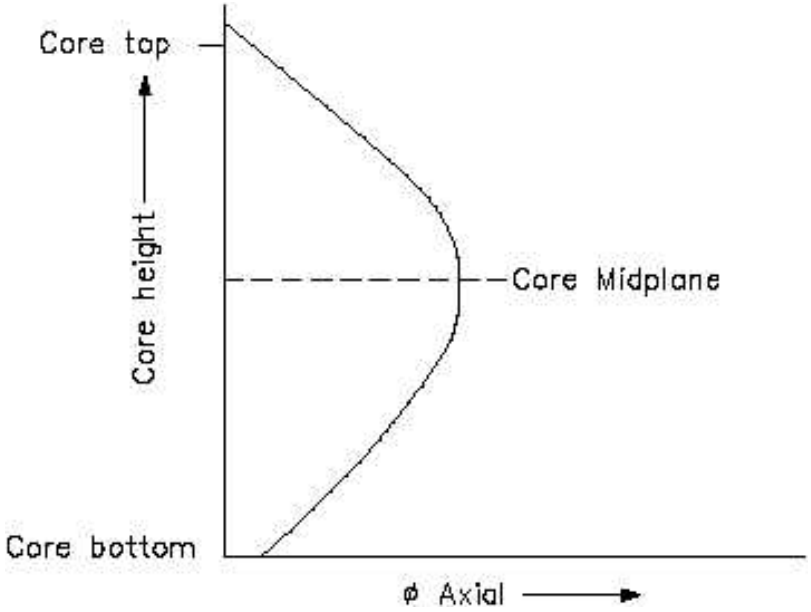

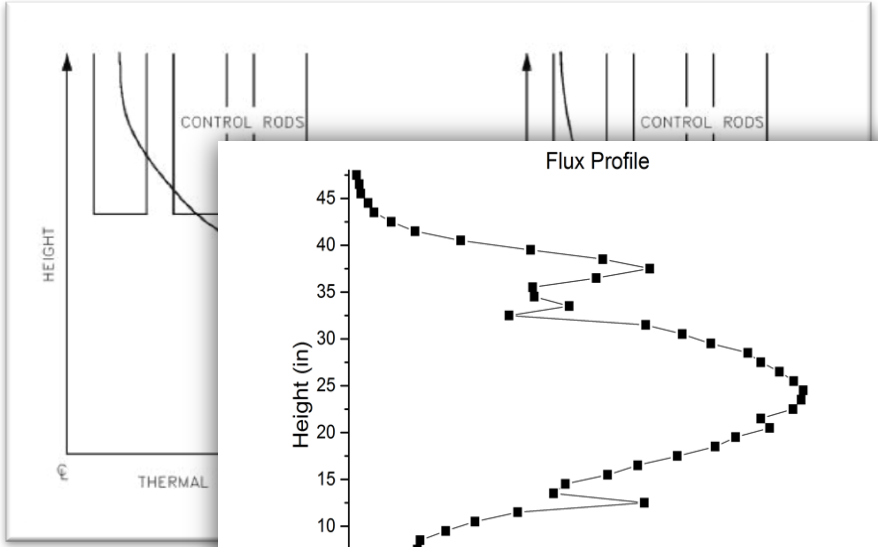

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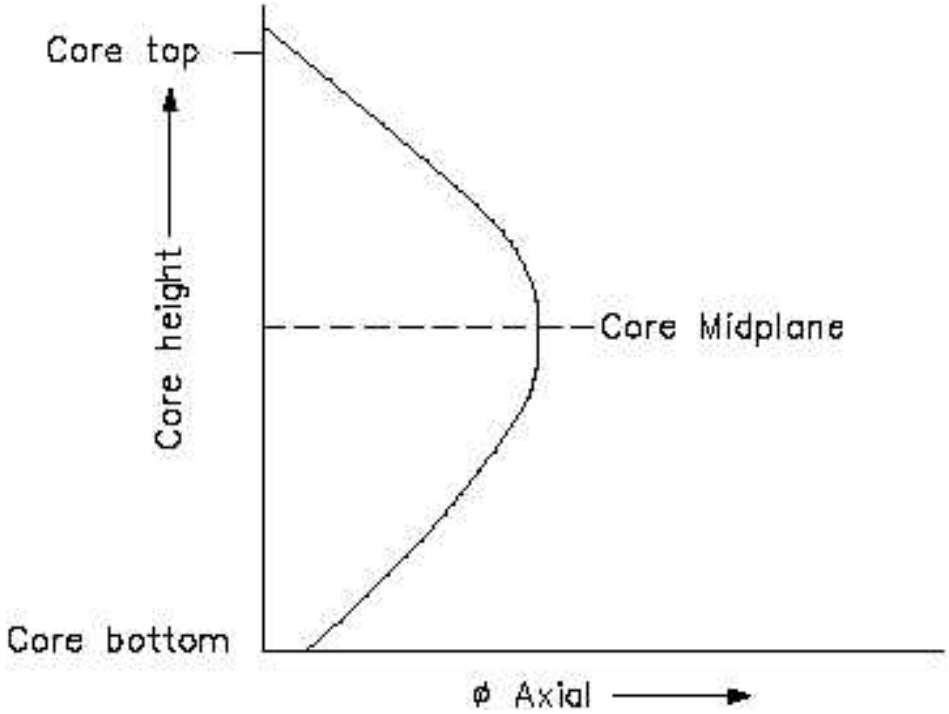
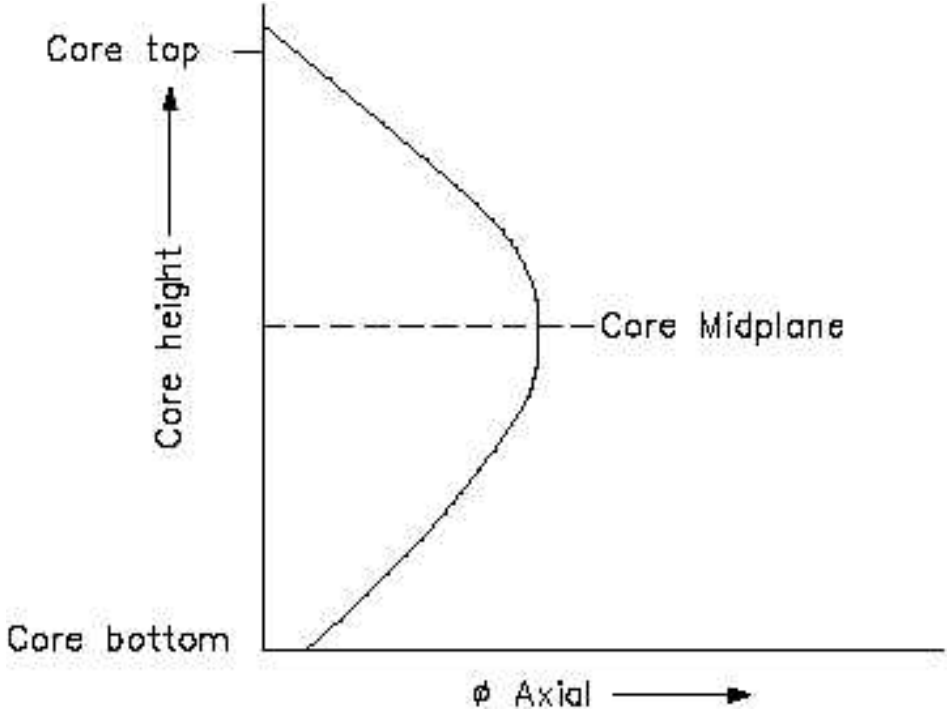


Sources: U.S. Department of State (top), Johannes Gutenberg University (bottom)



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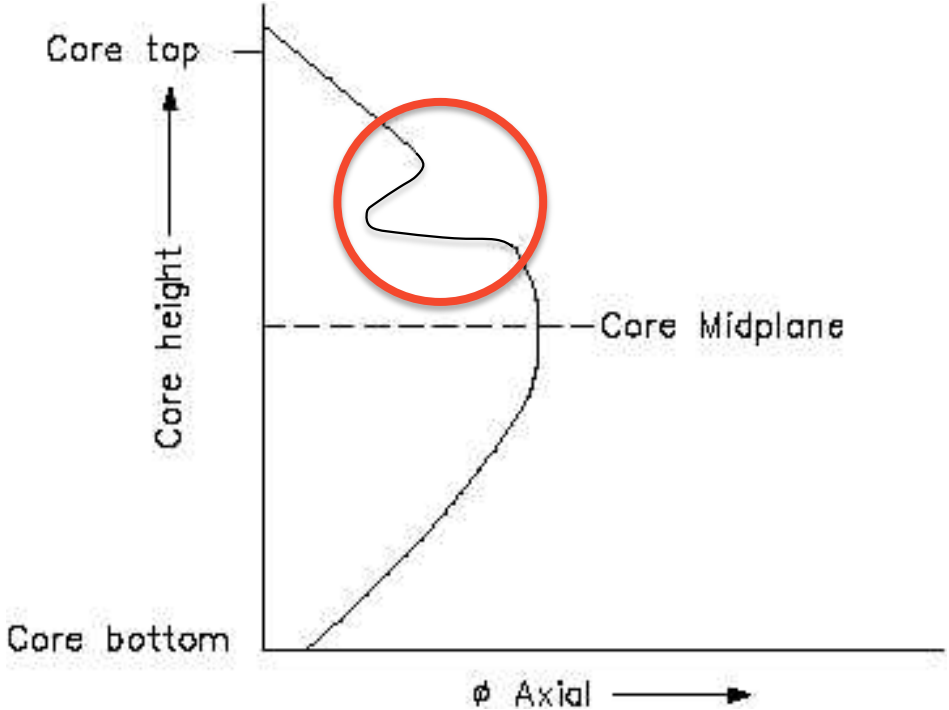

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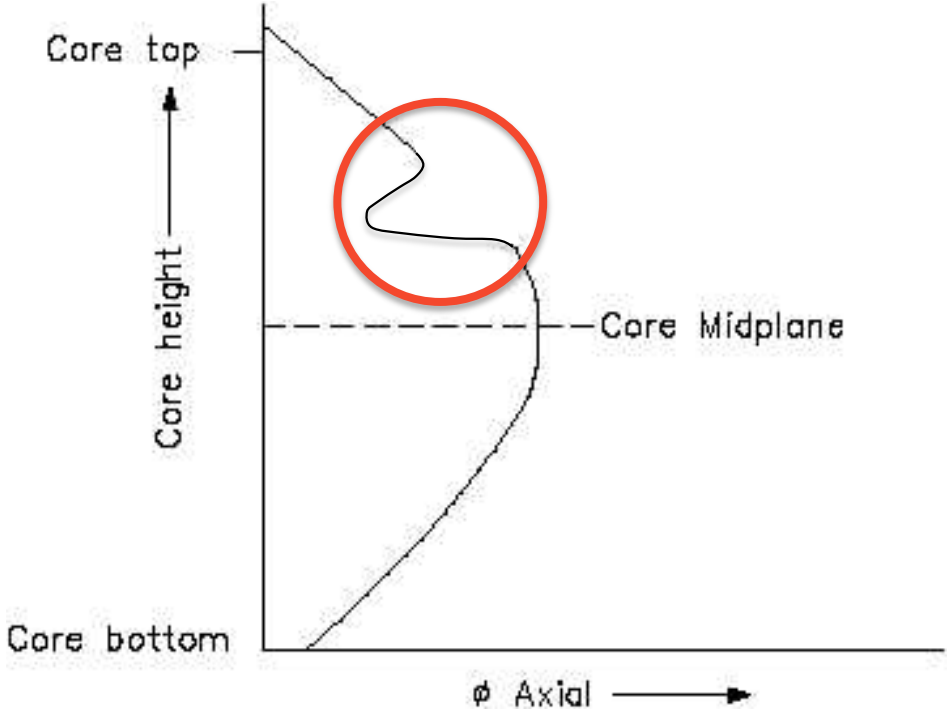

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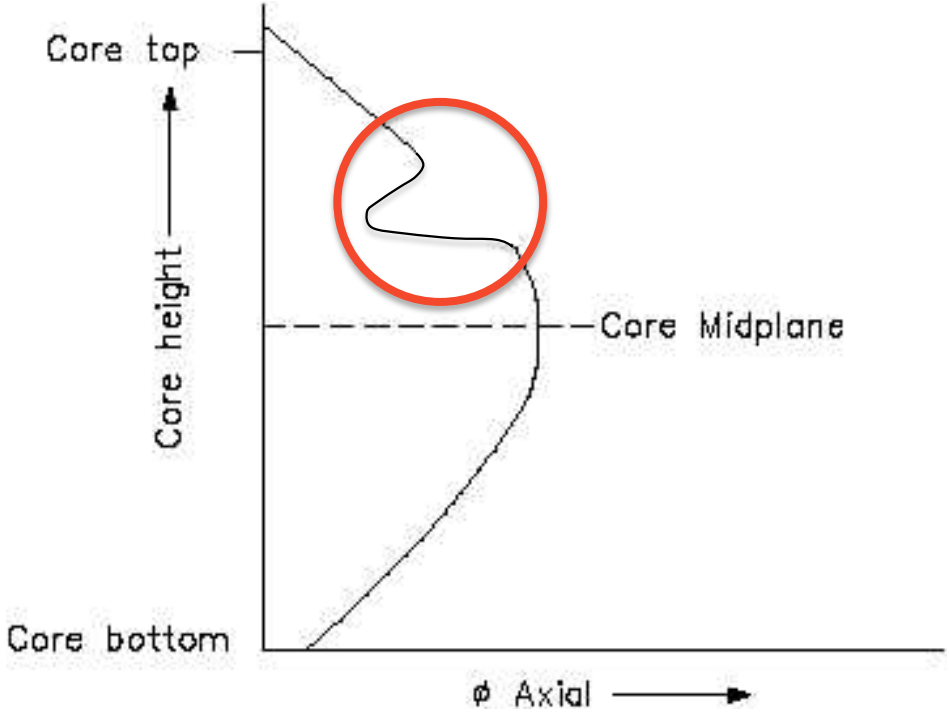

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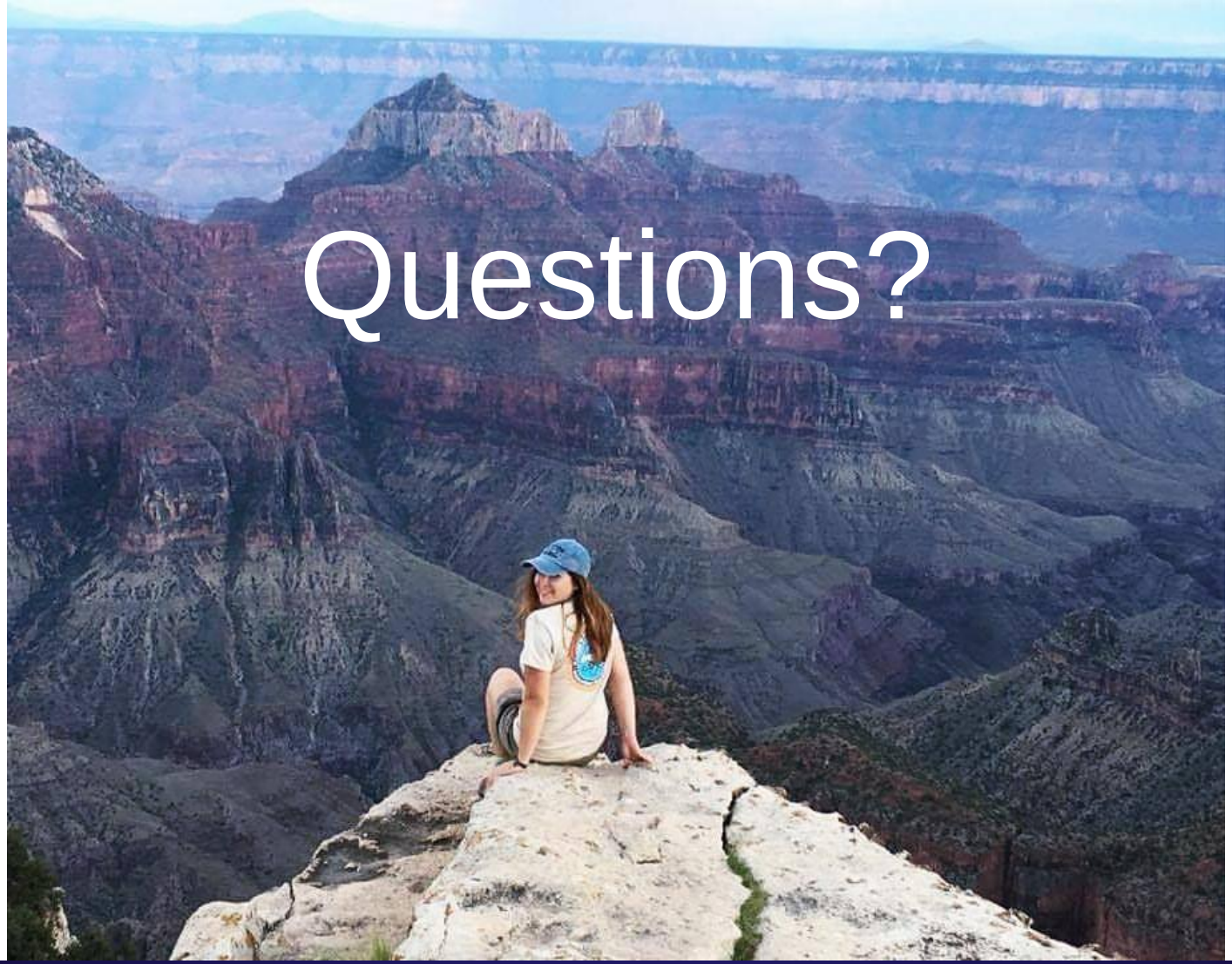
Weapons Grade Pu clad in Stainless Steel and surrounded by a Tungsten Reflector

Rail radiation portal monitor (RPM) at the Port of Antwerp, Belgium


Questions?